

# Optimal Paradigms: A challenge from Judeo-Tripolitanian Arabic\*

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## Abstract

This paper reevaluates Optimal Paradigms (OP), an extension to Optimality Theory proposed by McCarthy (2005). In OP, candidates are full inflectional paradigms and faithfulness constraints can require uniformity between all members within a paradigm. McCarthy (2005) proposed that OP can replace the assumption that prosodic templates are grammatical entities and other phonological mechanisms, which he regards as stipulations. One of the case studies used in McCarthy (2005) to showcase OP comes from Moroccan Arabic, where a noun-verb asymmetry in the distribution of schwa is claimed to be predicted by paradigmatic constraints as a means to achieve uniformity within an inflectional paradigm. We reevaluate this proposal using new fieldwork data from Judeo-Tripolitanian Arabic, an endangered variety of Arabic that is closely related to Moroccan. We show that if adjectives are added to the picture, OP can no longer predict the distribution of schwa in the different categories. The conclusion is that OP cannot replace the mechanisms it was supposed to replace, and is therefore a redundant complication to phonological theory.

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# 1 Introduction

Since the introduction of Correspondence Theory (McCarthy and Prince 1995) within Optimality Theory (OT; Prince and Smolensky 1993/2004), faithfulness constraints have been extended to refer not just to input-to-output correspondence (or to correspondence between a reduplicant and its base), but also to an additional correspondence relation that holds between outputs of related words. This output-to-output correspondence relation, along with the faithfulness constraints that refer to it, has been primarily proposed to account for cases where the phonology of morphologically-complex words applies opaquely, typically making those words more similar to their subconstituents. Theories assuming output-to-output correspondence go under names like Transderivational Faithfulness, Output-to-Output Correspondence, Base-Derivative Correspondence, and Paradigm Uniformity (e.g., Kenstowicz 1996; Benua 1997; Steriade 1999; Kager 1999; Albright 2010).

Within the output-to-output correspondence literature, two kinds of correspondence relations have been proposed. The first is an asymmetric relation between a ‘base+affix’ form and its base. The second is a symmetric correspondence relation that holds between each pair of members in a paradigm. The focus of this paper is on Optimal Paradigms (OP; McCarthy 2005), a theory that assumes the symmetric kind of correspondence relation between outputs.

In OP, candidates are full inflectional paradigms and faithfulness constraints can require uniformity between all members within a candidate. OP has not been proposed to replace other correspondence mechanisms. Indeed, Albright (2010) has shown that OP does not eliminate the necessity of the asymmetric kind of output-to-output correspondence. Instead, the addition of OP by McCarthy was intended to replace other theoretical mechanisms, which he regards as stipulations.

As presented by McCarthy, OP is a theory that can help advance the goals of a program he calls “Generalized Templatic Theory”. This program tries to eliminate any theoretical mechanism proposed specifically for deriving so-called templatic effects, which are recurring prosodic shapes of a group of morphemes. Per McCarthy, “OP supports the minimalist goals of Generalized Template Theory, which seeks to eliminate templates and similar stipulations from linguistic theory, replacing them with independently motivated constraints” (McCarthy, 2005, p. 171). Like previous authors who have responded to McCarthy (for example, Bobaljik 2008 and Smith 2011), we understand “similar stipulations” to include the phonological reference to morphosyntactic categories, which has been previously used to account for category-specific phonological patterns. Throughout his paper, McCarthy showcases OP using case studies of phonological asymmetry between nouns and verbs, justifying his new OP analyses by saying that “the difference between nouns and verbs is what we seek to explain, not to stipulate” (McCarthy, 2005, p. 199). Bobaljik (2008) has named this approach “*the thesis of category-neutral phonology*” (TCNP).

According to the TCNP, systematic differences between categories are solely epiphenomenal and should be derived without direct reference to the categories themselves. At first sight, the TCNP seems to be challenged by many patterns described in the litera-

ture as involving systematic phonological differences between categories (see, among others, Smith 2016). McCarthy (2005) uses a pattern from Classical Arabic to illustrate. In Classical Arabic, verbal stems are restricted to the templates in (1) and must end with a CVC syllable. Examples include fa.ʔal, faʔ.ʔal, fa.ʔal, tfaʔ.ʔal, tfa.ʔal, nfa.ʔal, jaf.ʔal, ju.faʔ.ʔil, ju.fa.ʔil, ja.ta.faʔ.ʔal, ja.ta.fa.ʔal, and others.

- (1) Possible verbal templates: stems with a final CVC syllable.

$$(C) \left\{ \begin{array}{l} CV \\ CVC \\ CV: \end{array} \right\}.CVC$$

Nouns, on the other hand, have no such restriction and their stems can end with either CVC, CV:C, or CVCC syllables. Examples include fa.ʔal, fa.ʔa:l, faʔl, dah.raḥ, dah.ra:h, and others.

McCarthy (2005) proposed that this noun-verb asymmetry in Classical Arabic can be predicted by OP, by appealing to the different paradigm structures of verbs and nouns. We will introduce OP in detail in section 2, but here is a simplified version of McCarthy's analysis. The general phonology of Classical Arabic determines that stems, regardless of category, cannot end with a CV:C syllable before C-initial suffixes (e.g., \*[fa.ʔa:l.tu]). A markedness constraint that penalizes tri-moraic syllables ( $*\mu\mu\mu]_\sigma$ ) can motivate shortening before such suffixes (/faʔa:l.tu/ → [faʔal.tu]). Simultaneously, an OP constraint that requires uniformity across the entire paradigm can motivate shortening before V-initial suffixes as well, despite the fact that such forms do not violate the markedness constraint that motivates shortening. The logic of this proposal is schematically illustrated in (2).

$$(2) \quad \begin{array}{ccc} \text{fa.ʔa:l.tu} & \xrightarrow[\text{Uniformity}]{\text{Markedness}} & \text{fa.ʔal.tu} \\ \text{fa.ʔa:l.la} & \xrightarrow{\text{Uniformity}} & \text{fa.ʔa.la} \end{array}$$

Nouns in Classical Arabic, on the other hand, have no C-initial suffixes and therefore can end with any kind of syllable. A stem-final CVCC syllable can be similarly ruled out in verbs, but not in nouns, using OP constraints and without templates or direct reference to categories.

Bobaljik (2008) has already pointed out a weakness in supporting OP with this and similar cases. Since there is a correlation between the morphosyntactic category and paradigm structure in Classical Arabic, the noun-verb asymmetry can be accounted for by either OP constraints or category-specific constraints, templates, etc. Therefore, such cases do not provide evidence for OP.<sup>1</sup>

<sup>1</sup>Bobaljik further challenged OP using a pattern from Itelmen. He argued that this pattern cannot be accounted for without the phonology making direct category reference. However, as Bobaljik has already noted, Itelmen does not provide a knock-down argument against OP. It has also been shown by Cable (2004), that OP can rise to the Itelmen challenge under certain assumptions and when combined with other OT extensions. Other than Itelmen, category-specific phonology is often reported in the phonological literature. For example, Smith (2011) has examined multiple cases of category-distinct phonological behaviour (including

We reevaluate one of McCarthy’s (2005) original case studies provided to support OP, involving the distribution of schwa in Moroccan Arabic. This case study is particularly interesting because it has been argued by McCarthy to manifest an OP prediction he named “Majority Rules”, which we will discuss in section 2. Briefly, Majority Rules is expected when markedness constraints rule out a completely uniform paradigm and cause discrepancies between groups of paradigm members. If, in such scenario, some paradigm members can satisfy these markedness constraints by aligning with either group, OP predicts they will align with the group consisting of more members.

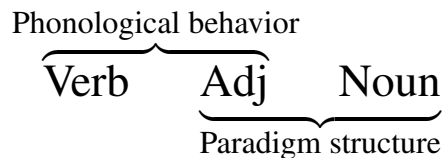
Turning to the Moroccan Arabic case study, tri-consonantal verbs in the language always surface as CCəC but tri-consonantal nouns are realized as either CCəC or CəCC depending on the relative sonority of the consonants. Specifically, the noun surfaces as CəCC if the second consonant in the noun is more sonorous than the third (in accordance with the Sonority Sequencing Generalization (SSG; Selkirk 1984)). Only otherwise the noun is realized as CCəC, the shape observed consistently in verbs. This distinct behavior can lead to minimal pairs across categories, as in (3).

- (3) A noun-verb minimal pair in Moroccan Arabic (McCarthy, 2005)  
 ʃərb ‘drinking’ ~ ʃərb ‘he drank’

McCarthy (2005) proposed that the different position of schwa in (3) is not determined by category-specific constraints that refer to the categories “noun” and “verb” directly. Instead, he suggested that the distinct distribution of schwa across categories is predicted by OP using the different paradigm structures of verbs and nouns.

In this paper, we challenge McCarthy’s (2005) analysis using fieldwork data from Judeo-Tripolitanian Arabic, an endangered variety of Arabic that is closely related to the Moroccan variety that McCarthy (2005) analyzes. In Judeo-Tripolitanian Arabic, the distribution of schwa in verbs and nouns follows the same generalizations as in Moroccan Arabic. We will see, however, that if we also take adjectives into consideration, paradigm structure can no longer predict the distinct behavior of the three different categories: schwa is distributed in adjectives as in verbs, but contradictory to OP, adjectives share the paradigm structure of nouns. As summarized in (4), the distribution of schwa in adjectives challenges OP because it defies the OP prediction that category-distinct behavior should align with paradigm structure. Judeo-Tripolitanian Arabic therefore disputes the original motivation for OP, which was illustrated using a similar pattern in Moroccan Arabic.

- (4) A schematic summary of the challenge from Judeo-Tripolitanian Arabic




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Itelmen) and suggested that some of them cannot be explained without phonological reference to categories. Smith’s survey, however, does not present a further challenge to OP beyond Itelmen.

The paper is structured as follows. In section 2 we introduce OP and present the motivation for the theory using McCarthy’s case study from Moroccan Arabic, which constitutes necessary background for our argument. In section 3 we present the new case study from Judeo-Tripolitanian Arabic. First, we show data from nouns and verbs that reflect the same generalizations that hold in nouns and verbs in Moroccan, and then, we present the data from adjectives that go beyond McCarthy’s (2005) examples. Finally, we show how the data contradict the predictions of OP.

## 2 OP and the distribution of schwa in Moroccan Arabic

There has been extensive research on the distribution of schwa in Moroccan Arabic (see Kaye 1987; Benhallam 1989/1990; Boudlal 2006/2007, 2011; Bensoukas and Boudlal 2012). It is established that schwa is always located between  $C_2$  and  $C_3$  in tri-consonantal verbs, which surface as  $CC\textcircled{a}C$  (e.g., [ʃrəb] ‘he drank’, [ktəb] ‘he wrote’). Nouns of the same form, however, surface as  $C\textcircled{a}CC$  if  $C_2$  is more sonorous than  $C_3$  (e.g., [kəlɓ] ‘dog’, [bórd] ‘wind’, [dénd] ‘sin’). Only otherwise, the noun surfaces as  $CC\textcircled{a}C$  (e.g., [rɜəl] ‘leg’, [ktəf] ‘shoulder’, [ħbəl] ‘rope’).<sup>2</sup> This distinct behaviour leads to noun-verb minimal pairs when  $C_2 > C_3$  in sonority, as shown in (3) and repeated in (5).

- (5) A noun-verb minimal pair in Moroccan Arabic (McCarthy, 2005)

ʃərb ‘drinking’ ~ ʃrəb ‘he drank’

Previous analyses accounted for this asymmetry by referencing nouns or verbs directly. Kaye’s 1987 analysis assumes different templates for nouns and verbs, as well as category-specific phonological rules. Boudlal (2006/2007) proposed noun-specific sonority constraints within OT. Within a framework that obeys the TCNP, however, such analyses are impossible. McCarthy therefore disregards past accounts of the pattern, suggesting that “OP can solve a long-standing problem in the phonology of Moroccan Arabic” (McCarthy, 2005, p. 202). He proposes a category-independent analysis that explains the Moroccan Arabic asymmetry using what he assumes are the different inflectional paradigms of the two categories, given in (6).

- (6) Inflectional suffixes in Moroccan Arabic according to McCarthy (2005)

| Verbs     |                   | Nouns     |           |
|-----------|-------------------|-----------|-----------|
| V-Initial | C-initial         | V-Initial | C-initial |
| əɪ, -u    | -t, -na, -ti, -tu | ∅         | ∅         |

The list in (6) consists of the inflectional suffixes of the verbal perfect paradigm.<sup>3</sup> Per

<sup>2</sup>Both verbs and nouns surface as  $C\textcircled{a}CC$  if they end with a geminate (e.g., [ʃədd] ‘to hold’, [dəmm] ‘blood’; Boudlal 2006/2007). In any analysis, there could be a constraint that motivates this position of schwa if a form ends with a geminate. Because nouns and verbs do not differ from each other in this respect, forms with a geminate are not going to be relevant to the discussion in this paper.

<sup>3</sup>For simplicity, McCarthy (2005) does not include the imperfective verb in his demonstration. However, he notes that his analysis should work either way.

McCarthy (2005), nouns do not have inflectional suffixes, an assumption we will return to later. In Moroccan Arabic's verbal perfect paradigm, schwa is located between  $C_2$  and  $C_3$  in the bare form ([rəb] 'he drank') and in stems that precede C-initial suffixes ([rəbt] 'I drank', [rəbna] 'we drank', [rəbti] 'you.sg drank', and [rəbtu] 'you.pl drank'). Schwa is located between  $C_1$  and  $C_2$  only in stems that precede V-initial suffixes ([ərbət] 'she drank' and [ərbu] 'they drank'). As we will see, this fact can make the noun-verb asymmetry predictable by OP, whose basic premises are presented in (7).

(7) The basic premises of Optimal Paradigms (McCarthy 2005)

- Candidates are full inflectional paradigms.
- All stems in a paradigm are in a correspondence relation with each other.
- In addition to markedness and input-to-output faithfulness constraints, there are output-to-output faithfulness constraints that require uniformity between corresponding members in a paradigm (OP constraints).

Using the properties in (7), an OP constraint can favor a verbal CCəC bare form but not a nominal CCəC form without making direct reference to categories. The idea will be that since most verbal suffixes are C-initial, and CCəC stems are better phonologically than CəCC before C-initial suffixes, a paradigm uniformity constraint can cause the bare stem to align with the majority of the paradigm and surface as CCəC as well. In addition to this OP constraint, McCarthy (2005) uses two markedness constraints and a cover constraint called SONCON that enforces the sonority effect observed in nouns.<sup>4</sup>

(8) Moroccan Arabic constraints (McCarthy, 2005)

- a. \*CCC: Assign \* for every sequence of three consonants.
- b. \*ə]<sub>σ</sub>: Assign \* for every schwa in an open syllable.
- c. OP-MAX-V: Assign \* for each pair of members in a paradigm in which one member has a vowel in a position where the other member does not.
- d. SONCON: Assign \* for a CCəC form if  $C_2 > C_3$  in sonority.

Both markedness constraints, \*CCC and \*ə]<sub>σ</sub>, reflect broad generalizations in the language. The first, \*CCC, penalizes sequences of three consonants. It is violated by forms such as \*[ərb-t] and can therefore motivate the occurrence of schwa between  $C_2$  and  $C_3$  before C-initial suffixes (/rb-t/ → [rəb-t]). The second constraint, \*ə]<sub>σ</sub>, penalizes schwas in open syllables. It is violated by forms such as \*[rəb-u] and can motivate the occurrence of schwa between  $C_1$  and  $C_2$  before V-initial suffixes (/rb-u/ → [ərb-u]). The OP constraint

<sup>4</sup>McCarthy (2005) does not formalize SONCON. However, a possible deconstruction of SONCON can be done using Boudlal's (2006/2007) constraints. Put simply, Boudlal suggests that a schwa must share a mora with a consonant, and proposes a hierarchy of constraints of the form \*əX]<sub>μ</sub>, where S = stop, F = fricative, N = nasal, L = liquid, and G = glide. These are ranked so as to favor a schwa before X over a schwa before X' if X is more sonorous than X'. Boudlal defines these constraints as noun-specific, in violation of the TCNP.

mentioned above, OP-MAX-V, requires uniformity between paradigm members with respect to the position of vowels. This constraint is violated twice by a pair of corresponding members, such as ⟨[rəb-t, fərb-u]⟩, in which the first member has a schwa where the second does not, and vice versa. This OP constraint can therefore motivate a schwa between C<sub>2</sub> and C<sub>3</sub> in the suffix-less form [rəb], although the alternative [fərb] does not violate any markedness constraints. The logic of this proposal is schematized in (9).<sup>5</sup>

$$\begin{array}{lll}
 (9) & \text{[rb-t]} & \xrightarrow[\text{*CCC}]{\text{Markedness}} \text{[rəbt]} \\
 & \text{[rb-u]} & \xrightarrow[\text{*ə}]_{\sigma}^{\text{Markedness}} \text{[fərbu]} \\
 & \text{[rb]} & \xrightarrow[\text{OP-MAX-V}]{\text{Uniformity}} \text{[rəb]}
 \end{array}$$

To see how the analysis works, consider the tableau in (10). In this tableau, candidates (10c) and (10d) have uniform paradigms (with CCəC and CəCC stems respectively), and therefore do not violate OP-MAX-V. However, since \*CCC and \*ə]<sub>σ</sub> outrank OP-MAX-V, both candidates are eliminated: candidate (10c) has two forms with a schwa in an open syllable (the paradigm members with V-initial suffixes), while candidate (10d) has four forms with a tri-consonantal sequence (the paradigm members with C-initial suffixes).

(10) An OP analysis of the distribution of schwa in Moroccan Arabic verbs

|                                                                                        | /[rb/ + {t, na, ti, tu, ət, u}]                          | *ə] <sub>σ</sub> | *CCC  | OP-MAX-V | SONCON |
|----------------------------------------------------------------------------------------|----------------------------------------------------------|------------------|-------|----------|--------|
| a.  | ⟨[rəb, frəbt, frəbna, frəbti<br>frəbtu, fərbət, fərbu]⟩  |                  |       | 20 *'s   | *      |
| b.                                                                                     | ⟨[fərb, frəbt, frəbna, frəbti<br>frəbtu, fərbət, fərbu]⟩ |                  |       | 24 *'s!  |        |
| c.                                                                                     | ⟨[rəb, frəbt, frəbna, frəbti<br>frəbtu, fərbət, fərbu]⟩  | *!*              |       |          | *      |
| d.                                                                                     | ⟨[fərb, fərbt, fərbna, fərbti<br>fərbtu, fərbət, fərbu]⟩ |                  | *!*** |          |        |

The remaining two non-uniform candidates, (10a) and (10b), satisfy both markedness constraints, and differ only in the position of schwa in the suffix-less member ([rəb] in (10a) vs. [fərb] in (10b)). This difference is reflected in the number of violations of OP-MAX-V. Since (10a) has five CCəC stems and two CəCC stems, there is a total of 20 violations of OP-MAX-V (5 · 2 pairs of ⟨CəCC, CCəC⟩ · 2 violations for each pair). In (10b), on the other hand, there are four CCəC stems and three CəCC stems, which incur a total of 24 violations of OP-MAX-V (4 · 3 · 2). Candidate (10a) is therefore selected as the optimal paradigm, even though it also violates the lower-ranked SONCON.

<sup>5</sup>This discussion assumes that schwas in Moroccan Arabic are epenthetic, following the studies of Benhallam (1989/1990), Boudlal (2006/2007), and Bensoukas and Boudlal (2012). However, as pointed out by McCarthy (2005), this assumption is not crucial because the analysis works with any of the underlying representations /fərb/, /fərb/, /fərb/, and /fərb/.

McCarthy (2005) treats this pattern as an attested case of Majority Rules, a special prediction of OP. Majority Rules occurs when alternative forms of a paradigm member equally satisfy the same markedness constraints (like [rób] and [jórb]). In such cases, OP predicts that the member will be aligned with the majority of other members. As emphasized in (11), the reason schwa is located between C<sub>2</sub> and C<sub>3</sub> in verbs like [rób] is because there are more members in the verbal paradigm with C-initial suffixes than V-initial suffixes. To the best of our knowledge, this is the only pattern that has been argued to exhibit Majority Rules.<sup>6</sup>

- (11) The Majority Rules prediction regarding the position of schwa in Moroccan Arabic  
If C<sub>1</sub>C<sub>2</sub>əC<sub>3</sub> and C<sub>2</sub> > C<sub>3</sub> in sonority ⇒ |C-initial| > |V-initial| inflectional suffixes.

To complete the analysis of Moroccan Arabic, the same paradigm-uniformity effect should not be obtained in nouns. To this end, as mentioned above, McCarthy proposes that Moroccan Arabic nouns do not have inflectional affixes: he asserts that possessive clitic pronouns are not part of the inflectional paradigm, and points out that broken plurals are not formed by regular suffixation and therefore should not be considered part of an inflectional paradigm. We will not evaluate this stipulation, as the question of what constitutes a paradigm can remain a black box while we test OP. As we will see later, OP will fail on the case study from Judeo-Tripolitanian Arabic regardless of any assumptions about what counts as part of the paradigm.

For now, however, under McCarthy's assumption, nominal candidates consist of only one member, the bare singular noun. As shown in (12), nouns therefore vacuously satisfy OP-MAX-V. Since neither a CəCC form nor a CCəC form violates the higher-ranked markedness constraints, the position of schwa is determined by SONCON.

- (12) An OP analysis of the distribution of schwa in Moroccan Arabic nouns

|                                                                                        | /[rb]/ | *ə] <sub>σ</sub> | *CCC | OP-MAX-V | SONCON |
|----------------------------------------------------------------------------------------|--------|------------------|------|----------|--------|
| a.                                                                                     | [rób]  |                  |      |          | *!     |
| b.  | [jórb] |                  |      |          |        |

The position of schwa in nouns can therefore be seen as an instance of the emergence of the unmarked (McCarthy and Prince, 1994), in which the sonority preference of Moroccan Arabic appears because nominal candidates consist of a single member and higher-ranked constraints are inactive. Per McCarthy (2005), this is not a noun-specific preference, but rather a category-independent preference that does not emerge in verbs because of their paradigm structure. Before turning to the challenge posed by adjectives, it is worth mentioning Caubet's (1993) generalization that collective nouns in Moroccan Arabic surface

<sup>6</sup>Majority Rules here should not be confused with a different phenomenon sometimes referred to in the literature as a Majority Rules effect (see, e.g., Baković 2000). The latter involves harmony patterns rather than a case where a majority of paradigm members exert their influence on other members.



as CCəC regardless of sonority, e.g., [brək] ‘duck’.col vs. [bgər] ‘cow’.col.<sup>7</sup> This generalization does not follow from McCarthy’s (2005) analysis, as some collective nouns violate SONCON. Therefore, they may pose an additional challenge to OP, which we will not pursue in this paper.

### 3 The challenge to OP from Judeo-Tripolitanian Arabic

#### 3.1 The distribution of schwa in JTA nouns and verbs

Judeo-Tripolitanian Arabic (henceforth, JTA) is a variety of Arabic that used to be spoken in Tripoli, the capital city of Libya. JTA is one of several varieties spoken today exclusively by Jews who emigrated to Israel from Arabic-speaking countries.<sup>8</sup> Yoda (2005) has laid the groundwork for research of this variety with an in-depth documentation and preliminary description of its morphophonology. His work was based on 35 hours of speech from nine different consultants, seven of whom immigrated to Israel from Tripoli in 1949-1952 and two in 1967. Yoda’s description provides many insights into the phonology and morphology of JTA and includes a significant portion of the crucial data that we will use below. However, we have verified the examples we will see in an independent field research, to make sure that the relevant data are all generated by the same individual grammar (see de Lacy 2014 for the importance of this point), and in order to create a publicly available audio database with high-quality recordings. As far as we know, the present work constitutes the first theoretical study of the variety.

Our own data include about 5,000 different words and sentences elicited from a single JTA native speaker, during over 25 hours of recordings. The speaker was born in Tripoli in 1947 and immigrated with his family to Israel in 1949. Most of the examples presented below can also be found in Yoda’s documentation and any datum found in both studies shows no discrepancies that are relevant to this paper.

Like in Moroccan Arabic, the distribution of schwa in verbs with trilateral roots is unconstrained by the sonority of the consonants.<sup>9</sup> This is shown in (13) in the perfect

<sup>7</sup>Kaye (1987) and Caubet (1993) also state that deverbal nouns (traditionally called ‘Masdar’ in Arabic) always surface as CəCC. However, their data do not consist of enough convincing examples, in which nouns surface as CəCC and  $C_2 \leq C_3$  in sonority.

<sup>8</sup>In Israel, *Judeo-Arabic* speakers were under pressure to abandon their mother tongue in favor of Hebrew, and therefore did not pass it on to younger generations (see Hary 2019 for a political and historical discussion). In 1943 there were approximately 30,000 Jews in Libya, most of whom emigrated to Israel between 1949 and 1952 (Goldberg, 1983). Today an estimated number of only a few hundred native speakers of JTA are under the age of 65, which makes this variety critically endangered.

<sup>9</sup>In verbs of quadrilateral roots the case is slightly different: while sonority does not play a role in word-final clusters, it does play a role in medial clusters. CCC medial clusters are allowed if in the root,  $C_2 > C_3$  in sonority (e.g., [fər]k-ət] ‘she freshened’). If  $C_2 < C_3$ , however, a schwa is inserted after  $C_2$  (e.g., [ʕáfəlk-ət] ‘she choked’). Note that a word such as [fər]k-ət] also violates McCarthy’s (2005) constraint \*CCC. Whatever makes quadrilateral verbs violate this constraint is not significant to this paper and will not be explored further.

paradigms of the verbs [ʃrób] ‘he drank’ where  $C_2 > C_3$  in sonority; [gdám] ‘he bit’, where  $C_2 < C_3$  in sonority; and [ktʃób] ‘he wrote’, where  $C_2 \simeq C_3$  in sonority.

(13) Perfect paradigms for [ʃrób], [gdám], and [ktʃób]

| Inflection | Suffix | ʃ.r.b   | g.d.m   | k.tʃ.b   |      |
|------------|--------|---------|---------|----------|------|
| 1 SG       | -tʃ    | ʃróbʃ   | gdámʃ   | ktʃóbʃ   | CCəC |
| PL         | -na    | ʃróbna  | gdámna  | ktʃóbna  |      |
| 2 MSG      | -tʃ    | ʃróbʃ   | gdámʃ   | ktʃóbʃ   |      |
| FSG        | -tʃi   | ʃróbʃi  | gdámʃi  | ktʃóbʃi  |      |
| PL         | -tʃu   | ʃróbʃu  | gdámʃu  | ktʃóbʃu  |      |
| 3 MSG      | ∅      | ʃrób    | gdám    | ktʃób    |      |
| FSG        | -ətʃ   | ʃərbətʃ | gədmətʃ | kətʃbətʃ | CəCC |
| PL         | -u     | ʃərbu   | gədmu   | kətʃbu   |      |

As can be seen in (13), regardless of the sonority of the consonants, schwas are located between  $C_2$  and  $C_3$  in the bare 3MSG form and before C-initial suffixes, but between  $C_1$  and  $C_2$  before V-initial suffixes. This is precisely the Moroccan distribution of schwa that we have seen in section 2.

The distribution of schwa in JTA nouns is also identical to Moroccan Arabic’s and, unlike verbs, is constrained by sonority. As can be seen in (14), McCarthy’s (2005) generalizations hold for JTA as well. Nouns surface as CCəC only if  $C_2 \leq C_3$  in sonority. If  $C_2 > C_3$  in sonority, nouns surface as CəCC.<sup>10,11</sup>

(14) JTA tri-consonantal nouns

- a.  $C_1əC_2C_3$  if  $C_2 > C_3$  in sonority:  
kólb ‘dog’, bəntʃ ‘girl’, wəld ‘boy’, səms ‘sun’, málh ‘salt’, tʃólʒ ‘ice’
- b.  $C_1C_2əC_3$  if  $C_2 \leq C_3$  in sonority:  
xʃəm ‘nose’, hbəq ‘basil’, lhəm ‘meat’, sdər ‘chest’, bsəl ‘onion’, fhəm ‘coal’

Up until now, the phonological behaviour of JTA has been exactly the same as the behavior McCarthy (2005) assumed for Moroccan Arabic. In what follows, however, we will see the distribution of schwa in adjectives, which is not discussed in McCarthy’s original analysis of Moroccan. Adjectives in JTA surface as CCəC regardless of sonority – just

<sup>10</sup>Like in Moroccan, both verbs and nouns in JTA surface as CəCC if  $C_2$  and  $C_3$  form a geminate (e.g., [ʃədd] ‘he grabbed’ and [fəmm] ‘mouth’). Again, this is not significant for us as there is no asymmetry between the categories in this respect.

<sup>11</sup>Stridency may also play a role in the positioning of schwa. Words such as [lós] ‘clothes’, [nəfs] ‘half’, and [kəbs] ‘horned male sheep’, in which  $C_2 \leq C_3$  in sonority but  $C_3$  is a strident, surface as CəCC. Since our relevant data only include these three examples, we refrain from making generalizations regarding stridents (however, Yoda 2005 has additional examples that suggest that this behavior of stridents in nouns is systematic). Apart from that, the only exception to McCarthy’s (2005) generalization that we have found is [hnəf] ‘snake’. Incidentally or not, this word has also been reported as an exception to the pattern in Moroccan Arabic (Boudlal, 2006/2007).

like verbs – but share the paradigm structure of nouns. In section 3.3 we will see how this fact creates a paradox for OP. Heath (2002) and Boudlal (2006/2007) state that adjectives in Moroccan Arabic also surface as CCəC, suggesting that the same challenge could have been posed using Moroccan Arabic if additional data from this variety had been available.

## 3.2 The distribution of schwa in JTA adjectives

JTA adjectives of the form that contains three consonants and a schwa correspond to adjectives of the form ?aCCaC in Classical Arabic. Many of them are from the adjectival category called *color/disability adjectives*,<sup>12</sup> which includes primarily colors and disabilities, but also other meanings. As can be seen in (15), the position of schwa in these adjectives is always between C<sub>2</sub> and C<sub>3</sub>, regardless of the sonority of the consonants.

### (15) Color/disability adjectives in JTA

- a. C<sub>2</sub> > C<sub>3</sub> in sonority:  
bjəd, ‘white’ zrəq ‘blue’, trəf ‘deaf’, hwəl ‘cross-eyed’
- b. C<sub>2</sub> < C<sub>3</sub> in sonority:  
sfər ‘yellow’, hmər ‘red’, khəl ‘black’, hlów ‘sweet’

Other adjectives of the same shape, which also correspond to ?aCCaC in Classical Arabic, are the comparative forms of JTA adjectives, exemplified in (16).

### (16) JTA comparatives

| Regular form | Comparative |         | Regular form | Comparative |          |
|--------------|-------------|---------|--------------|-------------|----------|
| qrəb         | qrəb        | ‘close’ | məzján       | zjén        | ‘pretty’ |
| kbír         | kbér        | ‘big’   | qsér         | qsér        | ‘short’  |
| smín         | smén        | ‘fat’   | tʃfóh        | tʃfóh       | ‘tasty’  |

As can be seen in (16), the distribution of schwa in comparatives is unconstrained by sonority as well. In [qrəb] and [zjén], for example, C<sub>2</sub> > C<sub>3</sub> in sonority, in [kbér] and [qsér] C<sub>2</sub> < C<sub>3</sub>, and in [smén] and [tʃfóh] C<sub>2</sub> = C<sub>3</sub>.

## 3.3 The challenge to OP from JTA adjectives

We have just seen the distribution of schwa in verbs, nouns, and adjectives separately from each other. The table in (17) shows the three categories side by side. In Adjectives, like in verbs but unlike in nouns, schwa is positioned between C<sub>2</sub> and C<sub>3</sub> if C<sub>2</sub> > C<sub>3</sub> in sonority.

<sup>12</sup>This category is sometimes inappropriately called *color/defect*.

(17) The distribution of schwa across categories

| Verbs |                        | Adjectives |                  | Nouns |                |                                                |
|-------|------------------------|------------|------------------|-------|----------------|------------------------------------------------|
| CCəC  |                        |            |                  | CəCC  |                | C <sub>2</sub> > C <sub>3</sub><br>in sonority |
| frób  | <i>‘he drank’</i>      | bjód       | <i>‘white’</i>   | kálb  | <i>‘dog’</i>   |                                                |
| tlób  | <i>‘he asked’</i>      | zróq       | <i>‘blue’</i>    | bóntʃ | <i>‘girl’</i>  |                                                |
| drób  | <i>‘he hit’</i>        | tróf       | <i>‘deaf’</i>    | wóld  | <i>‘boy’</i>   |                                                |
| qlóq  | <i>‘he was fed up’</i> | qrób       | <i>‘closer’</i>  | sóms  | <i>‘sun’</i>   |                                                |
| CCəC  |                        |            |                  |       |                | C <sub>2</sub> ≤ C <sub>3</sub><br>in sonority |
| gdóm  | <i>‘he bit’</i>        | sfér       | <i>‘yellow’</i>  | xfám  | <i>‘nose’</i>  |                                                |
| ktʃób | <i>‘he wrote’</i>      | kħál       | <i>‘black’</i>   | sdór  | <i>‘chest’</i> |                                                |
| xsól  | <i>‘he washed’</i>     | qsér       | <i>‘shorter’</i> | ħbér  | <i>‘basil’</i> |                                                |
| ħsób  | <i>‘he thought’</i>    | smón       | <i>‘fatter’</i>  | lhóm  | <i>‘meat’</i>  |                                                |

Recall McCarthy's (2005) OP analysis of Moroccan Arabic, in which category-free constraints, embodied by SONCON, favor CəCC over CCəC if  $C_2 > C_3$  in sonority. This is the sonority configuration in which the categories behave differently from each other. According to McCarthy's analysis, the reason that CəCC forms appear only in nouns is the Majority Rules prediction observed in verbs. According to his proposal, an OP constraint that outranks SONCON requires uniformity between paradigm members. This OP constraint forces the suffix-less verb to align with the majority of members of the verbal paradigm, which have C-initial suffixes. As was highlighted in (11), the OP prediction is therefore that if a stem surfaces as CCəC when  $C_2 > C_3$  in sonority, that stem must appear before more C-initial than V-initial inflectional suffixes.

While this Majority Rules prediction is consistent with the paradigm structures of verbs and nouns, this is not the case for JTA adjectives. As can be seen from the inflection of [tróf] 'deaf' in (18), JTA's full adjectival paradigm only consists of V-initial suffixes.

(18) The inflection of [tróf] 'deaf'

| Inflection | SR     |      |
|------------|--------|------|
| MSG        | tróf   | CCəC |
| FSG        | tərfá  | CəCC |
| PL         | tərfín |      |

The suffix-less form is realized as CCəC, even though the adjectival stem does not appear before *any* C-initial suffixes.

As shown in the tableau in (19), the suffix-less, adjectival form [tróf] not only violates the sonority constraints, but also the requirement of paradigm uniformity. Candidate (19a), which represents the actual forms in JTA, violates OP-MAX-V four times. This is because this candidate includes two ⟨CəCC, CCəC⟩ pairs of paradigm members in which the position of schwa is not uniform, and each of these pairs violates OP-MAX-V twice. Therefore, if the inflectional forms in (18) count as a paradigm, OP fails to generate the correct output.

- (19) OP fails on the distribution of schwa in JTA adjectives

|      | /trf/ + {a, in}       | *ə] <sub>σ</sub> | *CCC | OP-MAX-V | SONCON |
|------|-----------------------|------------------|------|----------|--------|
| a. ☹ | ⟨trəf, tərɣá, tərɣín⟩ |                  |      | *!***    | *      |
| b. ☹ | ⟨tərf, tərɣá, tərɣín⟩ |                  |      |          |        |

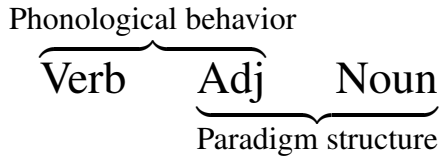
Recall McCarthy's (2005) stipulation that noun inflections do not constitute a paradigm for the purpose of OP. A similar stipulation for JTA adjectives will not resolve the challenge, as shown in (20).

- (20) OP fails on the distribution of schwa in JTA adjectives (assumption: adjectives have no inflectional paradigm)

|      | /trf/ | *ə] <sub>σ</sub> | *CCC | OP-MAX-V | SONCON |
|------|-------|------------------|------|----------|--------|
| a. ☹ | trəf  |                  |      |          | *!     |
| b. ☹ | tərf  |                  |      |          |        |

If adjectival paradigms consist of only one suffix-less member and the sonority effect on schwa placement is indeed category-independent, the prediction is that the effect would emerge in adjectives as well. However, as can be seen in (20), SONCON favors the wrong candidate (20b) since  $C_2 > C_3$  in sonority ( $r > f$ ). Therefore, regardless of what constitutes a paradigm for JTA adjectives, the distribution of schwa in adjectives contradicts the prediction that category-distinct behaviour is determined by paradigm structure. As summarized in (4) and repeated here in (21), while adjectives behave like verbs phonologically, they share the paradigm structure of nouns – a contradiction from the perspective of OP.

- (21) A schematic summary of the challenge from Judeo-Tripolitanian Arabic



Accounts of the distribution of schwa in JTA can be easily developed, however, as long as the phonology can make use of prosodic templates and direct reference to morphosyntactic categories. For demonstration, consider the constraint in (22), which is a templatic constraint in the spirit of Bat-El's (2003) constraints for Semitic morphology. Consider also the tableau in (23), which illustrates a possible derivation of /trf/ using this constraint.

- (22)  $\{CCəC\}_{\text{Verb, Adj}}$ : The stem of a verb or an adjective must surface as CCəC.

- (23) An account of the JTA pattern using category-specific templates

|      | /trf/ <sub>Adj</sub> | *ə] <sub>σ</sub> | *CCC | $\{CCəC\}_{\text{Verb, Adj}}$ | SONCON |
|------|----------------------|------------------|------|-------------------------------|--------|
| a. ☹ | trəf                 |                  |      |                               | *      |
| b. ☹ | tərf                 |                  |      | *!                            |        |

Because /trf/ is an adjective, it must satisfy  $\{CC\bar{a}C\}_{Verb, Adj}$  and surface as  $CC\bar{a}C$ . This templatic constraint outranks SONCON and therefore candidate (23a) is chosen even though it violates the general sonority preference of the language. Verbs can be derived similarly as they are also required to satisfy  $\{CC\bar{a}C\}_{Verb, Adj}$ . Nouns, on the other hand, do not incur violations of  $\{CC\bar{a}C\}_{Verb, Adj}$  and therefore their form is determined by the lower-ranking SONCON. Given the ranking in (23), the position of schwa in suffixed forms is determined by the higher-ranking markedness constraints  $*\bar{a}]_{\sigma}$  and  $*CCC$  (as in McCarthy’s analysis). An alternative account is similarly possible if there are no templatic constraints, but the sonority preference is noun-specific and therefore only affects nouns, as proposed by Boudlal (2006/2007). Crucially, these solutions require direct reference to categories, in violation of the TCNP: the templatic analysis uses templatic constraints that make reference to “verb” and “adjective”, and the alternative uses a version of SONCON that makes reference to “noun”. Category-specific constraints can account not just for the JTA pattern, but also for the noun-verb asymmetry in McCarthy’s (2005) Classical Arabic pattern discussed above. Since OP cannot account for the JTA pattern without category-specific constraints, and since such constraints are sufficient to account for the original case studies that motivated the theory, they make OP a redundant complication to phonological theory.

## 4 Conclusion

We have seen a case study that challenges the original motivation for OP, a theory intended to predict category-distinct behavior from paradigm structure rather than from category-specific phonology. Reevaluating one of McCarthy’s (2005) core case studies, we have shown that once additional data are considered, the OP analysis breaks down if categories cannot be used. The Majority Rules prediction, singled out by McCarthy as support for his proposal, is in fact false for the data McCarthy used to demonstrate this prediction with. Our conclusion is therefore that even under the assumptions of OP, phonology should be able to refer to the categories themselves in order to explain systematic phonological differences between categories.

As mentioned in the introduction, OP cannot replace other output-to-output correspondence theories either (see Albright 2010). Taken together with our conclusion, there is little to no evidence that can support the addition of a complicated mechanism like OP in the first place. However, since OP was introduced, a few studies have proposed that it plays a role in grammar (see, for example, Pons-Moll’s 2013 analysis of Majorcan Catalan, and Haddad and Wiltshire’s 2014 analysis of Lebanese Arabic). Such studies raise a question that we have not answered in this reply, which is whether there are other sources of evidence that support the addition of OP to phonological theory.

## References

- Albright, Adam. 2010. Base-driven leveling in Yiddish verb paradigms. *Natural Language & Linguistic Theory* 28:475–537.
- Baković, Eric. 2000. Harmony, Dominance and Control. Doctoral Dissertation, Rutgers University, New Jersey.
- Bat-El, Outi. 2003. The fate of the consonantal root and the binyan in Optimality Theory. *Recherches Linguistiques de Vincennes* 32:31–60.
- Benhallam, Abderrafi. 1989/1990. Moroccan Arabic syllable structure. *Revue Langues et Littératures* 8:177–191.
- Bensoukas, Karim, and Abdelaziz Boudlal. 2012. The prosody of Moroccan Amazigh and Moroccan Arabic: Similarities in the phonology of schwa. In *Prosody matters: essays in honor of Elisabeth Selkirk*, ed. Tony Borowsky, Shigeto Kawahara, Mariko Sugahara, and Takahito Shinya, 3–42. London: Equinox.
- Benua, Laura. 1997. Transderivational identity: Phonological relations between words. Doctoral Dissertation, University of Massachusetts Amherst.
- Bobaljik, Jonathan. 2008. Paradigms (optimal and otherwise): A case for scepticism. In *Inflectional identity*, ed. Asaf Bachrach and Andrew Nevins, 29–54. Oxford University Press.
- Boudlal, Abdelaziz. 2006/2007. Sonority-driven schwa epenthesis in Moroccan Arabic. *Languages and Linguistics* 18&19:59–81.
- Boudlal, Abdelaziz. 2011. On the special behavior of schwa in Moroccan Arabic prosody. From the 8th Old World Conference in Phonology, Marrakech.
- Cable, Seth. 2004. Phonological noun-verb dissimilarities in Optimal Paradigms. Paper presented at Workshop on (Non)-identity within a paradigm, MIT.
- Caubet, Dominique. 1993. *L'arabe marocain*. Paris: Peeters.
- Goldberg, Harvey E. 1983. Language and culture of the Jews of Tripolitania: A preliminary view. *Mediterranean language review* 1:85–102.
- Haddad, Youssef A., and Caroline Wiltshire. 2014. Paradoxical paradigms! evidence from Lebanese Arabic phonology. In *Perspectives on Arabic linguistics XXVI*, ed. Karen Froud and Reem Khamis-Dakwar, 185–209. John Benjamins B.V.
- Hary, Benjamin. 2019. Judeo-Arabic in the Arabic-speaking world. In *Languages in Jewish communities, past and present*, ed. Benjamin Hary and Sarah Bunin Benor, 35–69. Berlin, Boston: De Gruyter Mouton.

- Heath, Jeffrey. 2002. *Jewish and Muslim dialects of Moroccan Arabic*. London: RoutledgeCurzon.
- Kager, René. 1999. *Optimality Theory*. Cambridge University Press.
- Kaye, Jonathan. 1987. Government in phonology: The case of Moroccan Arabic. *The Linguistic Review* 6:131–160.
- Kenstowicz, Michael. 1996. Base identity and uniform exponence: Alternatives to cyclicity. In *Current trends in phonology: Models and methods*, ed. Jacques Durand and Bernard Laks, 365–394. Salford: University of Salford Press.
- de Lacy, Paul. 2014. Evaluating evidence for stress systems. In *Word stress: Theoretical and typological issues*, ed. Harry van der Hulst, 149–193. Cambridge University Press.
- McCarthy, John, and Alan Prince. 1994. The emergence of the unmarked: Optimality in prosodic morphology. In *Proceedings of NELS 24*, ed. M. González, 333–379.
- McCarthy, John, and Alan Prince. 1995. Faithfulness and reduplicative identity. In *Papers in optimality theory*, ed. S. Urbanczyk J. Beckman L. Walsh Dickey, volume 18 of *University of Massachusetts Occasional papers*, 249–384. UMASS.
- McCarthy, John J. 2005. Optimal paradigms. In *Paradigms in phonological theory*, ed. Laura J. Downing, T. A. Hall, and Renate Raffelsiefen, 170–210. Oxford University Press.
- Pons-Moll, Clàudia. 2013. Underapplication of vowel reduction to schwa in Majorcan Catalan. some evidence for the left syllable of the stem as a prominent position and for subparadigms. In *proceedings of the 40th Annual Meeting of the North East Linguistic Society (NELS-40, 2009), MIT. GLSA (Graduate Linguistic Student Association)*, 121–135.
- Prince, Alan, and Paul Smolensky. 1993/2004. *Optimality Theory: Constraint interaction in generative grammar*. Wiley-Blackwell: Oxford.
- Selkirk, Elisabeth. 1984. On the major class features and syllable theory. In *Language sound structure: Studies in phonology presented to Morris Halle by his teacher and students*, ed. Mark Aronoff and Richard T. Oehrle, 107–136. Cambridge: MIT Press.
- Smith, Jeniffer L. 2016. Segmental noun/verb phonotactic differences are productive too. In *Proceedings of the Linguistic Society of America 1 (June)*, 16, 1–15.
- Smith, Jennifer L. 2011. Category-specific effects. In *The Blackwell companion to phonology*, ed. Marc van Oostendorp, Colin J. Ewen, Elizabeth V. Hume, and Keren Rice, 2439–2463. Malden, MA: Wiley-Blackwell.



Steriade, Donca. 1999. Lexical conservatism in French adjectival liaison. In *Formal perspectives in romance linguistics*, ed. B. Bullock, M. Authier, and L. Reed, 243–270. John Benjamins.

Yoda, Sumikazu. 2005. *The Arabic dialect of the Jews in Tripoli (Libya): grammar, text and glossary*, volume 35 of *Semitica viva*. Wiesbaden: Harrassowitz.